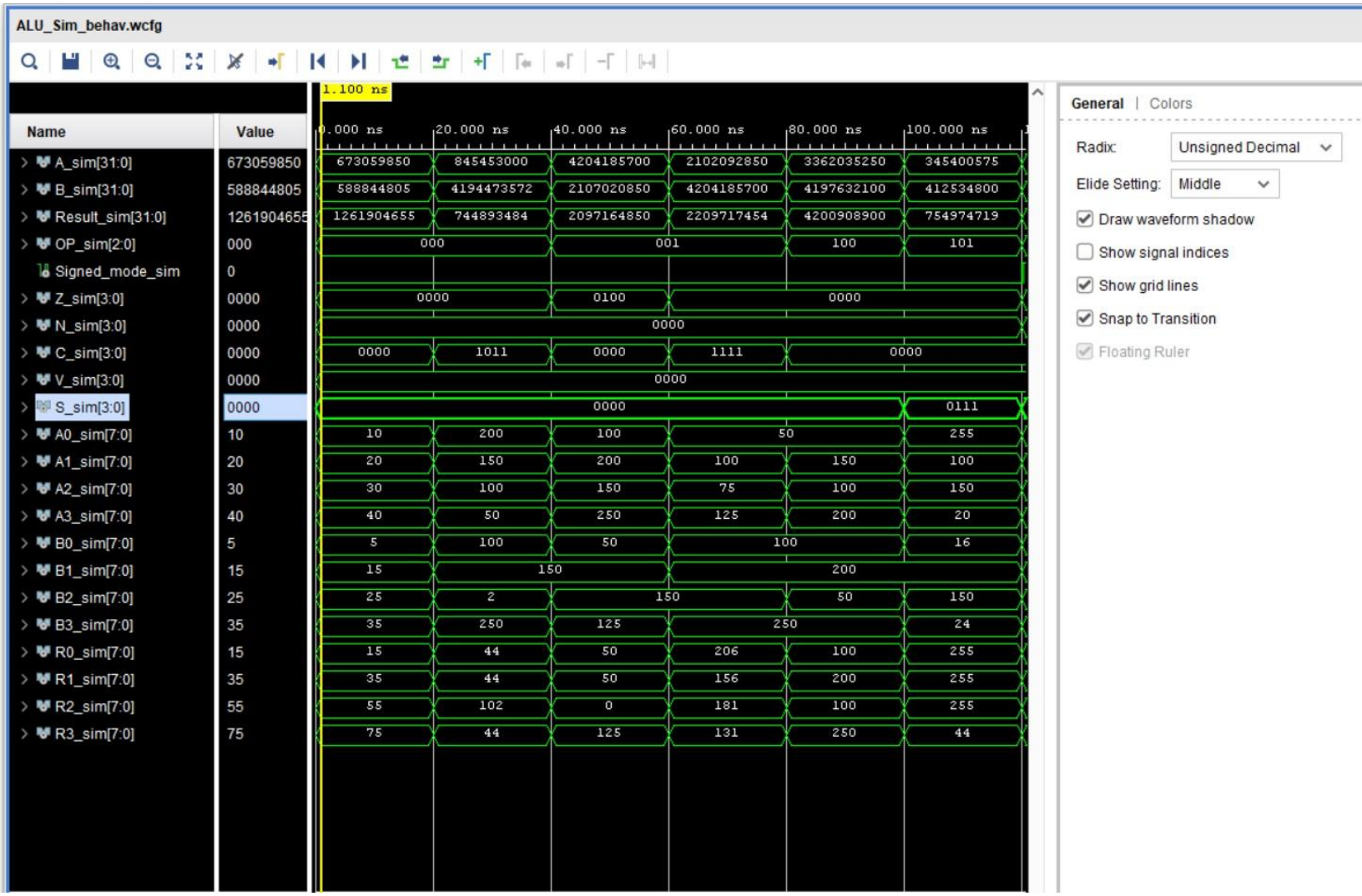
# Assignment 1 Report

## Question 3_1

**Behavioral and Post-Synthesis Timing Simulation:**

- In the simulation code, I write the expected results and the expected flags in the comments.
- The first photo is with an unsigned decimal radix (all unsigned tests are executed together), followed by all signed tests together. In the 3rd photo, the operations are bitwise, so for convenience, I set the radix to hexadecimal.



ALU_Sim_behav.wcfg

121.910 ns

| Name | Value | 120.000 ns | 140.000 ns | 160.000 ns | 180.000 ns | 200.000 ns | 220.000 ns |
|---|---|---|---|---|---|---|---|
| A_sim[31:0] | -585176054 | -585176054 | -2137234305 | -1671115214 | -1671135105 | -1671115214 | -2131360385 |
| B_sim[31:0] | 602345221 | 602345221 | -154258934 | -404285159 | 852364022 | -1253316839 | -164458230 |
| Result_sim[31:0] | 391951 | 391951 | 1986696841 | -1250052839 | 1788245641 | -1251678414 | -2144823681 |
| OP_sim[2:0] | 000 | 000 | | 001 | | 100 | 101 |
| Signed_mode_sim | 1 | | | | | | |
| Z_sim[3:0] | 1000 | 1000 | 0000 | | | | |
| N_sim[3:0] | 0010 | 0010 | 0011 | 1010 | 0101 | 1010 | |
| C_sim[3:0] | 0000 | 0000 | | | | | |
| V_sim[3:0] | 0000 | 0000 | 1111 | 0000 | 1111 | 0000 | |
| S_sim[3:0] | 0000 | 0000 | | | | | 1001 |
| A0_sim[7:0] | 10 | 10 | 127 | 50 | 127 | 50 | 127 |
| A1_sim[7:0] | -20 | -20 | 100 | -50 | -128 | -50 | 5 |
| A2_sim[7:0] | 30 | 30 | -100 | 100 | | | -10 |
| A3_sim[7:0] | -35 | -35 | -128 | -100 | | | -128 |
| B0_sim[7:0] | 5 | 5 | 10 | 25 | -10 | 25 | 10 |
| B1_sim[7:0] | 15 | 15 | 50 | 25 | 10 | -25 | -111 |
| B2_sim[7:0] | -25 | -25 | -50 | -25 | -50 | 75 | 50 |
| B3_sim[7:0] | 35 | 35 | -10 | -25 | 50 | -75 | -10 |
| R0_sim[7:0] | 15 | 15 | -119 | 25 | -119 | 50 | 127 |
| R1_sim[7:0] | -5 | -5 | -106 | -75 | 118 | -25 | -106 |
| R2_sim[7:0] | 5 | 5 | 106 | 125 | -106 | 100 | 40 |
| R3_sim[7:0] | 0 | 0 | 118 | -75 | 106 | -75 | -128 |

General | Colors

Radix: Signed Decimal

Elide Setting: Middle

- [x] Draw waveform shadow
- [ ] Show signal indices
- [x] Show grid lines
- [x] Snap to Transition
- [x] Floating Ruler

ALU_Sim_behav.wcfg

| Name | Value | 240.000 ns | 260.000 ns | 280.000 ns | 300.000 ns | 320.000 ns |
|---|---|---|---|---|---|---|
| A_sim[31:0] | dd1eec0a | ffaa550f | | 4080550f | | ffaa550f |
| B_sim[31:0] | 23e70f05 | 0f55aaff | | 00000000 | | 0f55aaff |
| Result_sim[31:0] | 0005fb0f | 0f00000f | f0fffff0 | 8000aale | | 072ad5ff |
| OP_sim[2:0] | 000 | 010 | 011 | 110 | | 111 |
| Signed_mode_sim | 1 | | | | | |
| Z_sim[3:0] | 1000 | 0110 | 0000 | 0100 | | 0000 |
| N_sim[3:0] | 0010 | 0000 | | 1010 | 0000 | |
| C_sim[3:0] | 0000 | 0000 | | | 0100 | 0000 |
| V_sim[3:0] | 0000 | 0000 | | 1110 | 0000 | |
| S_sim[3:0] | 0000 | 0000 | | | | |
| A0_sim[7:0] | 0a | 0f | | | | |
| A1_sim[7:0] | ec | 55 | | | | |
| A2_sim[7:0] | 1e | aa | | 80 | | aa |
| A3_sim[7:0] | dd | ff | | 40 | | ff |
| B0_sim[7:0] | 05 | ff | | 00 | | ff |
| B1_sim[7:0] | 0f | aa | | 00 | | aa |
| B2_sim[7:0] | e7 | 55 | | 00 | | 55 |
| B3_sim[7:0] | 23 | 0f | | 00 | | 0f |
| R0_sim[7:0] | 0f | 0f | f0 | 1e | | ff |
| R1_sim[7:0] | fb | 00 | ff | aa | | d5 |
| R2_sim[7:0] | 05 | 00 | ff | 00 | | 2a |
| R3_sim[7:0] | 00 | 0f | f0 | 80 | | 07 |

General | Colors

Radix: Hexadecimal

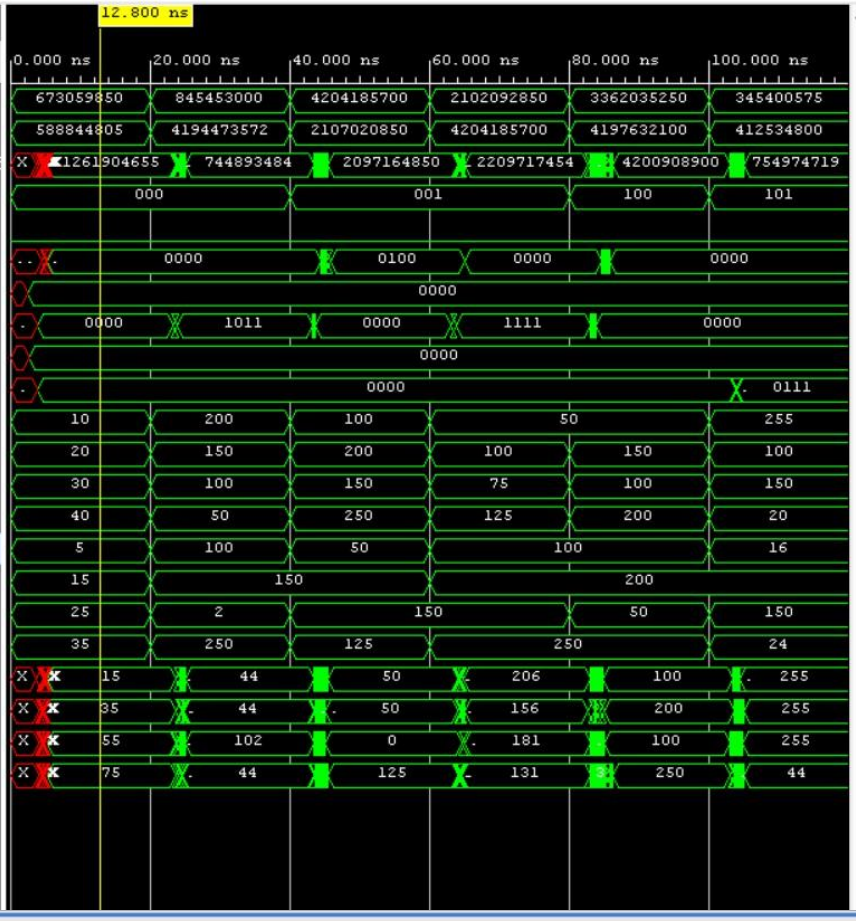
Elide Setting: Middle

- [x] Draw waveform shadow
- [ ] Show signal indices
- [x] Show grid lines
- [x] Snap to Transition
- [x] Floating Ruler

ALU_Sim_behav.wcfg

12.800 ns

| Name | Value |
|---|---|
| A_sim[31:0] | 673059850 |
| B_sim[31:0] | 588844805 |
| Result_sim[31:0] | 1261904655 |
| OP_sim[2:0] | 000 |
| Signed_mode_sim | 0 |
| Z_sim[3:0] | 0000 |
| N_sim[3:0] | 0000 |
| C_sim[3:0] | 0000 |
| V_sim[3:0] | 0000 |
| S_sim[3:0] | 0000 |
| A0_sim[7:0] | 10 |
| A1_sim[7:0] | 20 |
| A2_sim[7:0] | 30 |
| A3_sim[7:0] | 40 |
| B0_sim[7:0] | 5 |
| B1_sim[7:0] | 15 |
| B2_sim[7:0] | 25 |
| B3_sim[7:0] | 35 |
| R0_sim[7:0] | 15 |
| R1_sim[7:0] | 35 |
| R2_sim[7:0] | 55 |
| R3_sim[7:0] | 75 |




General | Colors

Radix: Unsigned Decimal

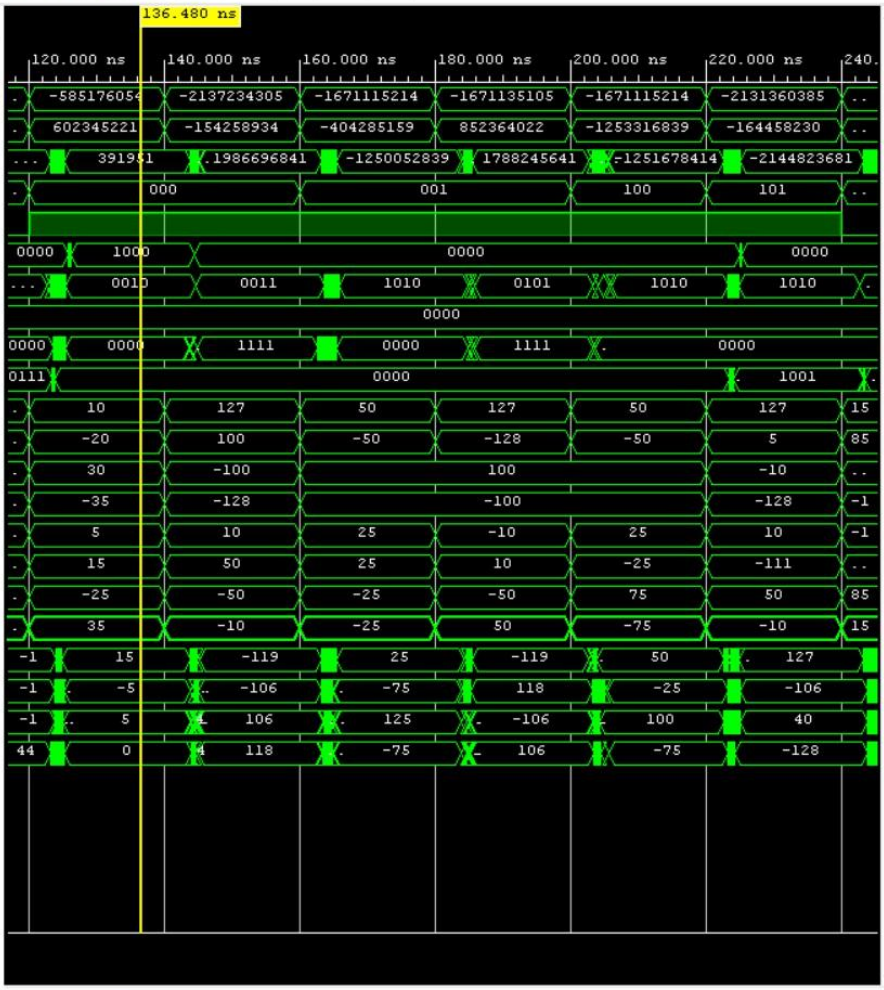
Elide Setting: Middle

- [x] Draw waveform shadow
- [ ] Show signal indices
- [x] Show grid lines
- [x] Snap to Transition
- [x] Floating Ruler

ALU_Sim_behav.wcfg

136.480 ns

| Name | Value |
|---|---|
| A_sim[31:0] | -585176054 |
| B_sim[31:0] | 602345221 |
| Result_sim[31:0] | 391951 |
| OP_sim[2:0] | 000 |
| Signed_mode_sim | 1 |
| Z_sim[3:0] | 1000 |
| N_sim[3:0] | 0010 |
| C_sim[3:0] | 0000 |
| V_sim[3:0] | 0000 |
| S_sim[3:0] | 0000 |
| A0_sim[7:0] | 10 |
| A1_sim[7:0] | -20 |
| A2_sim[7:0] | 30 |
| A3_sim[7:0] | -35 |
| B0_sim[7:0] | 5 |
| B1_sim[7:0] | 15 |
| B2_sim[7:0] | -25 |
| B3_sim[7:0] | 35 |
| R0_sim[7:0] | 15 |
| R1_sim[7:0] | -5 |
| R2_sim[7:0] | 5 |
| R3_sim[7:0] | 0 |

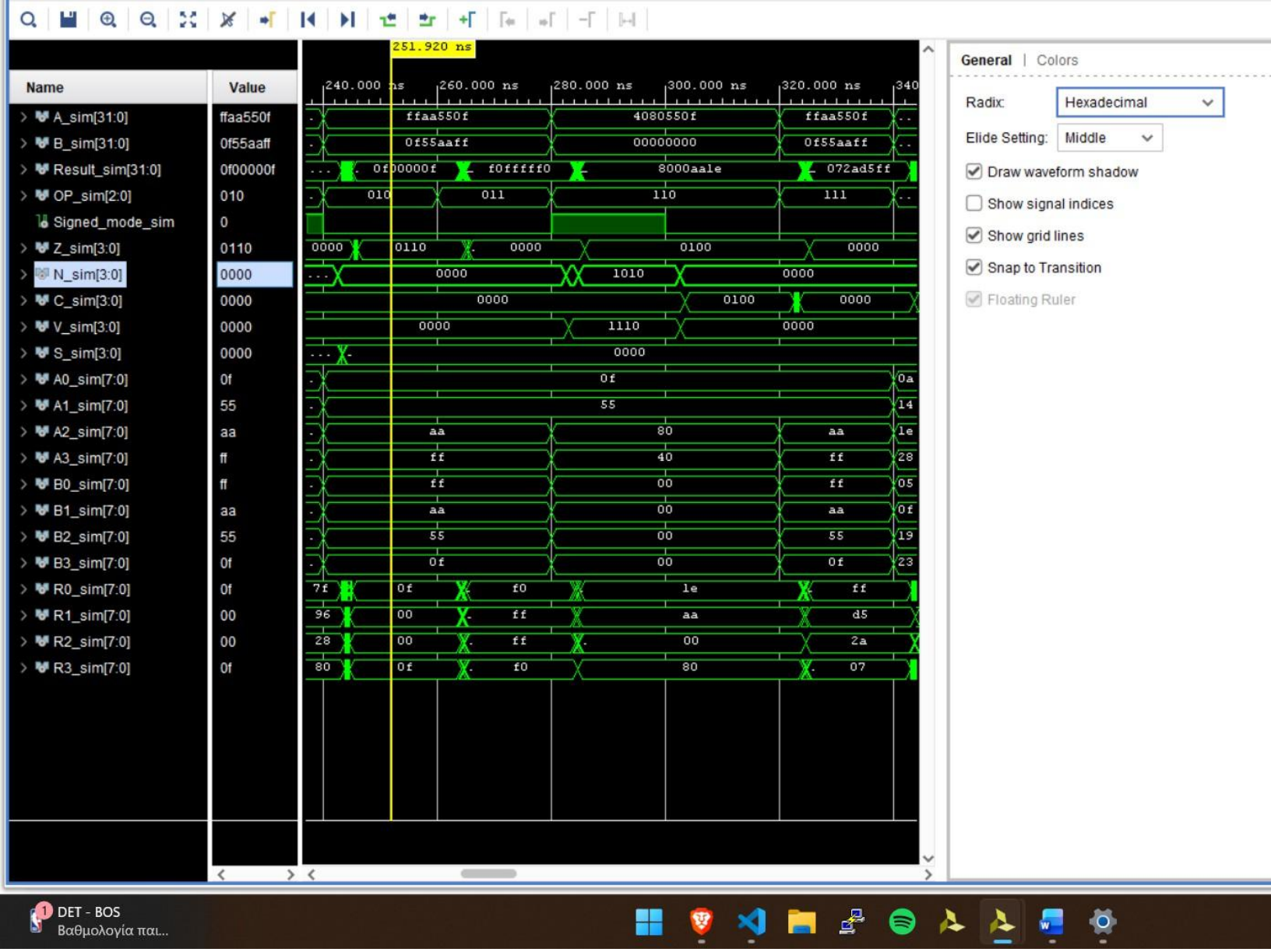



General | Colors

Radix: Signed Decimal

Elide Setting: Middle

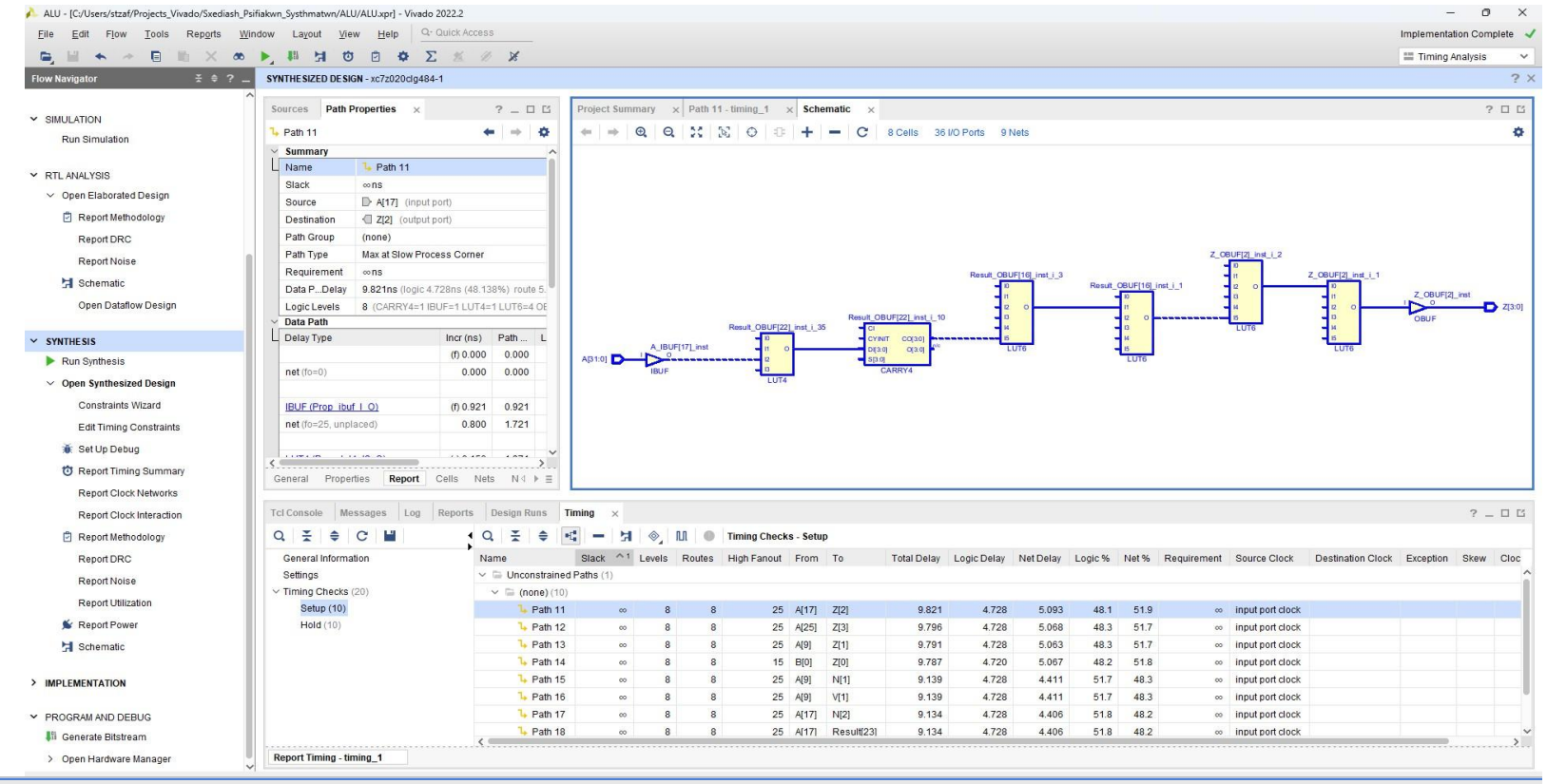
- [x] Draw waveform shadow
- [ ] Show signal indices
- [x] Show grid lines
- [x] Snap to Transition
- [x] Floating Ruler

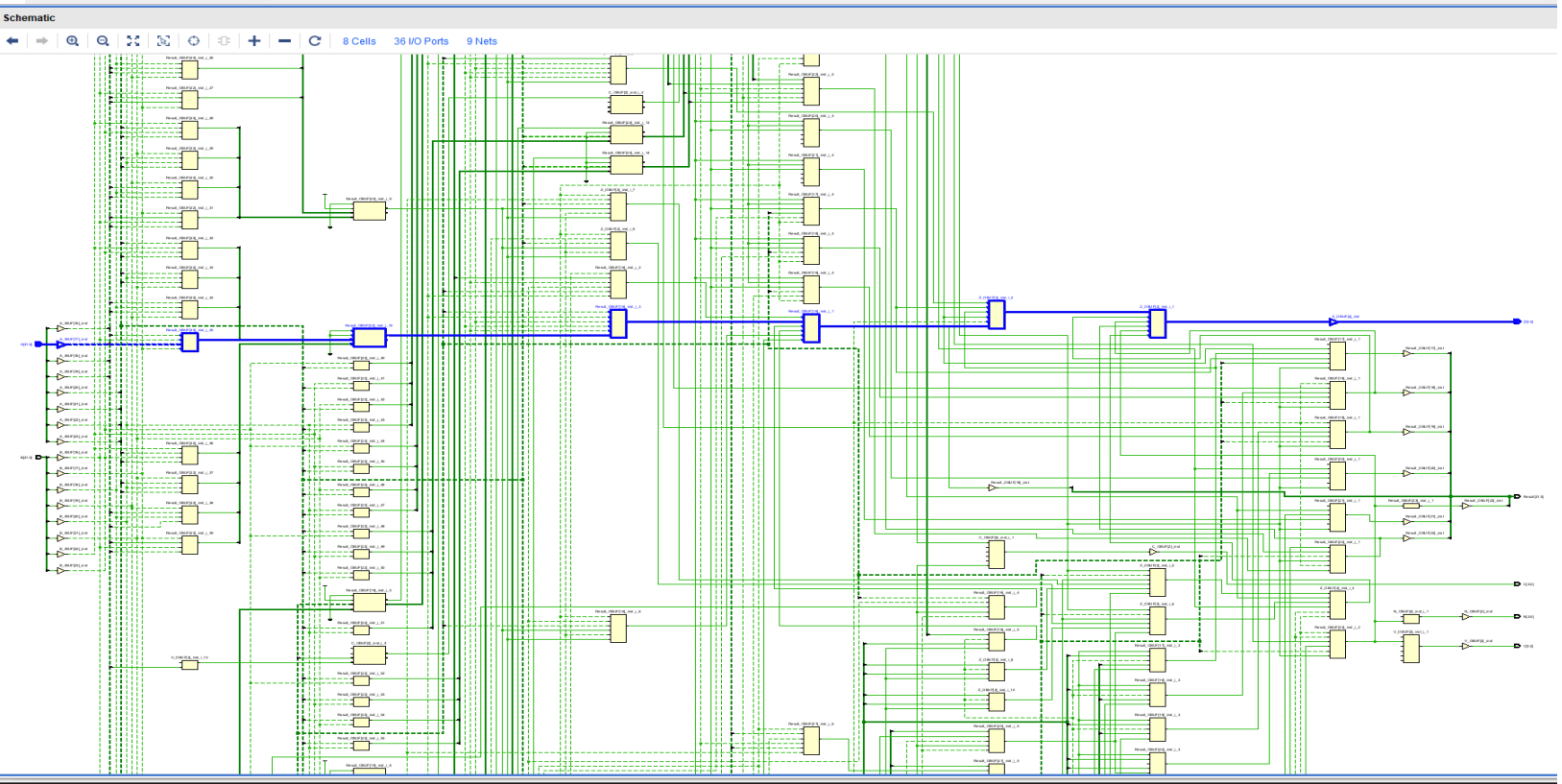



**The 3 last photos refer to Post synthesis timing Simulation**

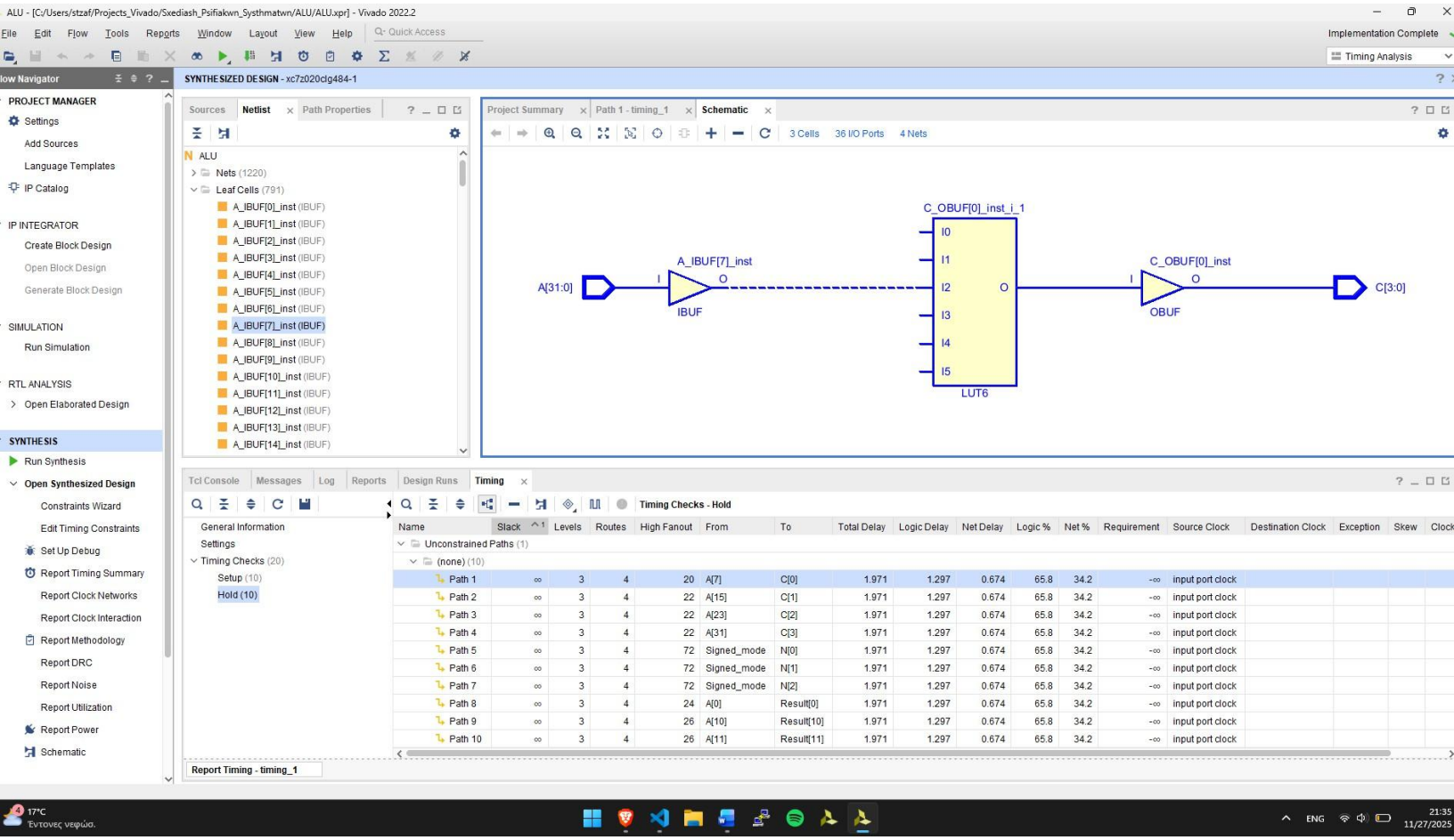
# Question 3_2

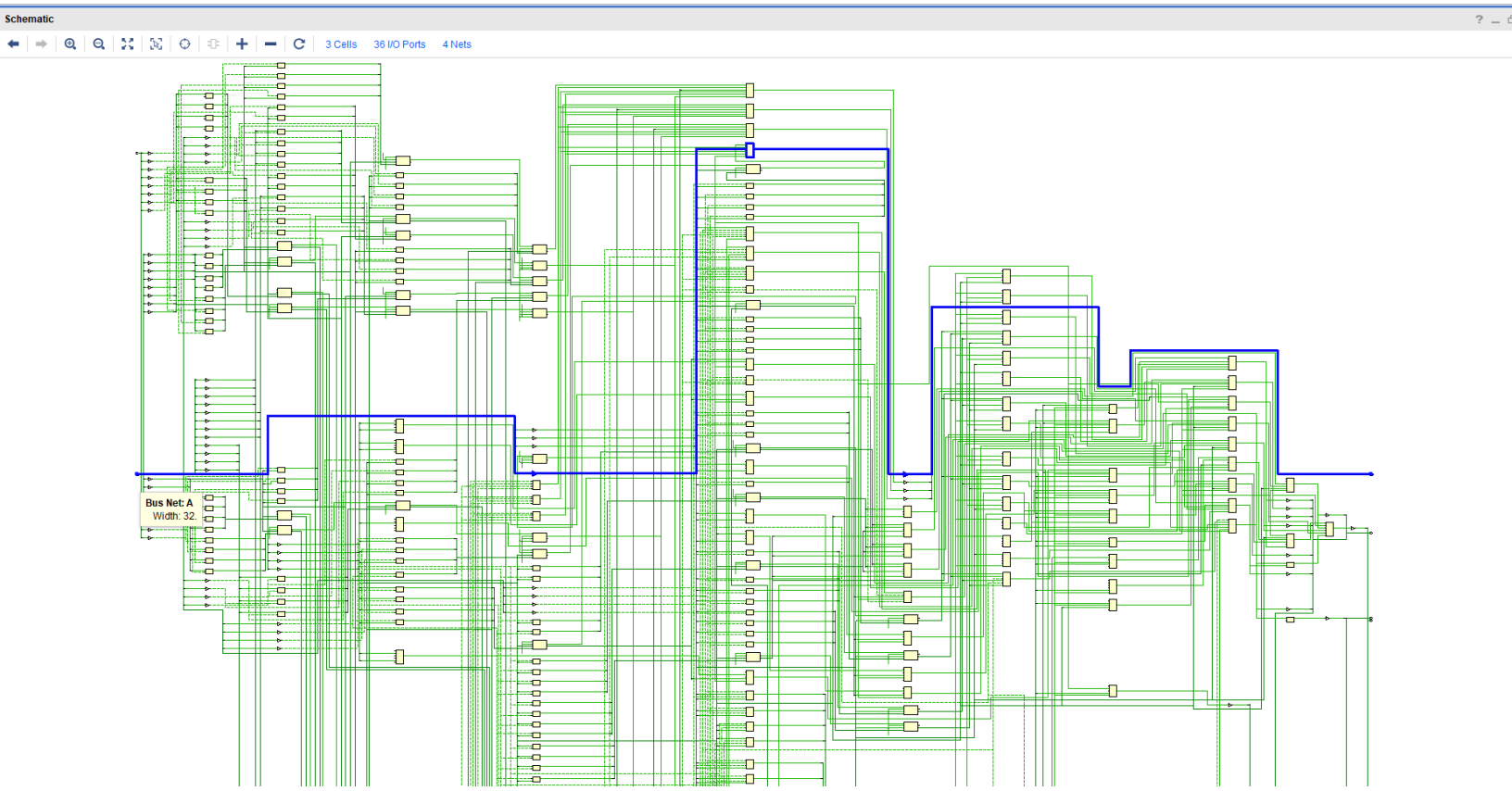
**Propagation Delay:** It is Path 11 – the slowest path (isolated and in the entire synthesized device).



**Contamination Delay:** All paths from 1 to 10 have the same minimum delay.




| Name | Slack | Levels | Routes | High Fanout | From | To | Total Delay | Logic Delay | Net Delay | Logic % | Net % | Requirement | Source Clock | Destination Clock | Exception | Skew | Clock |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Path 1 | ∞ | 3 | 4 | 20 | A[7] | C[0] | 1.971 | 1.297 | 0.674 | 65.8 | 34.2 | -∞ | input port clock | | | | |
| Path 2 | ∞ | 3 | 4 | 22 | A[15] | C[1] | 1.971 | 1.297 | 0.674 | 65.8 | 34.2 | -∞ | input port clock | | | | |
| Path 3 | ∞ | 3 | 4 | 22 | A[23] | C[2] | 1.971 | 1.297 | 0.674 | 65.8 | 34.2 | -∞ | input port clock | | | | |
| Path 4 | ∞ | 3 | 4 | 22 | A[31] | C[3] | 1.971 | 1.297 | 0.674 | 65.8 | 34.2 | -∞ | input port clock | | | | |
| Path 5 | ∞ | 3 | 4 | 72 | Signed_mode | N[0] | 1.971 | 1.297 | 0.674 | 65.8 | 34.2 | -∞ | input port clock | | | | |
| Path 6 | ∞ | 3 | 4 | 72 | Signed_mode | N[1] | 1.971 | 1.297 | 0.674 | 65.8 | 34.2 | -∞ | input port clock | | | | |
| Path 7 | ∞ | 3 | 4 | 72 | Signed_mode | N[2] | 1.971 | 1.297 | 0.674 | 65.8 | 34.2 | -∞ | input port clock | | | | |
| Path 8 | ∞ | 3 | 4 | 24 | A[0] | Result[0] | 1.971 | 1.297 | 0.674 | 65.8 | 34.2 | -∞ | input port clock | | | | |
| Path 9 | ∞ | 3 | 4 | 26 | A[10] | Result[10] | 1.971 | 1.297 | 0.674 | 65.8 | 34.2 | -∞ | input port clock | | | | |
| Path 10 | ∞ | 3 | 4 | 26 | A[11] | Result[11] | 1.971 | 1.297 | 0.674 | 65.8 | 34.2 | -∞ | input port clock | | | | |